Misc. Learning Paradigms

Varying dimensions:

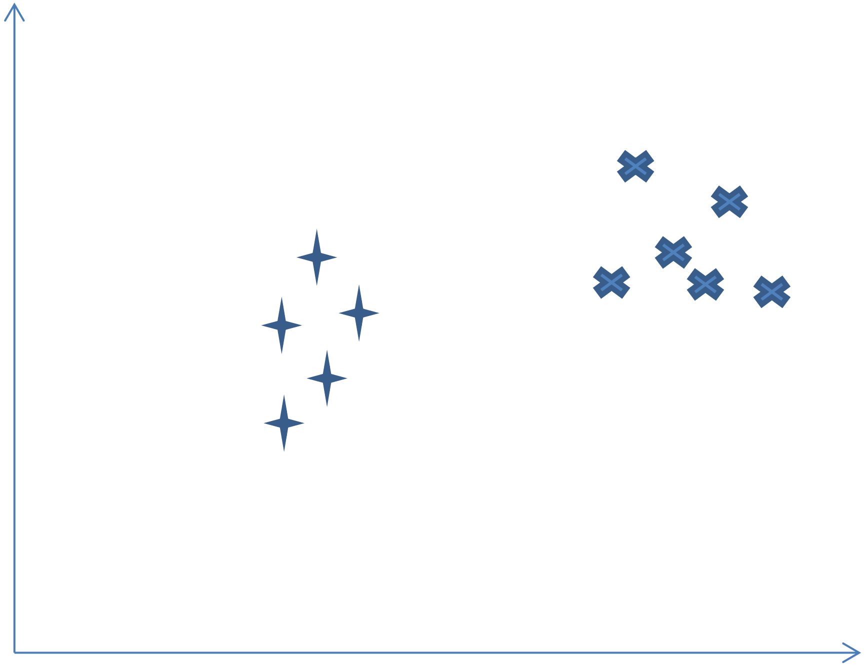
1) Amount of feedback (supervision)
2) Type of analysis
3) Types of Models
4) Types of algorithms

Amount of Feedback

Spectrum of feedback types, from least to most:

1. Completely unsupervised (e.g. clustering)
2. Reinforcement learning
3. Active learning
4. Supervised learning
5. Privileged learning

Clustering



Active Learning

Can no longer assume given training set of examples D with known X, Y values.

Many settings possible:

1) Example set D given only X values, learner may query value of Y on any of them, but at a cost.
2) Learner may generate examples and ask for classification (again at a cost).
3) Sometimes possible even during "testing", this is "online learning".

Reinforcement Learning

MDP version: we have an MDP,

but R and/or T unknown to the agent.

Recall Bellman equation:

U(s) = R(s) + gamma *max a (sum s’ U(s’)*T(s,a,s’))

Can re-write as:

U(s) = max a (R(s,a) + gamma * sum s’ U(s’)*T(s,a,s’))

= max a (Q(s,a))

Idea is to learn Q(s,a) by sampling action a multiple times, thereby also learning T(s,a,s’).

Supervised vs. Privileged

Standard supervised learning: training set D over attributes (X, Y), learn mapping M : X->Y

Privileged learning: additional data other than just values of attributes X, Y available during learning.

Most common case: training set D is over attributes (X, X*, Y), still learn mapping M : X->Y

Type of Analysis: Theoretical

Complexity of “learnability”.

Probably Approximately Correct (PAC)

Supervised learning version:

Given training set D with examples on attributes X, Y, drawn iid from distribution P

Learn mapping M:X -> Y, that will have error rate at most epsilon on test distributed as P,

in at least fraction 1-delta of the tries.

(Theoretical bounds on size of D needed, as a function of epsilon and delta, for certain types of P (concepts))

Type of Analysis: Empirical

How to empirically estimate learning performance?

We have seen: test set. Other methods:

- Validation sets
- Leave-one-out
- (X-fold) Cross validation

Measure several types of error: false positive, false negative, or combinations thereof.

In some cases can tune output “sensitivity”, can generate ROC curve and measure AUC.

Types of Models

Decision (and regression) trees

Logic formulae (special case: rules) (A^B^C=>D)

Neural Networks (including DL)

Bayes Networks

Support Vector Machines (SVM)

Automata (including stochastic automata)

Meta-models

- Bayesian Model Averaging
- Ensemble methods

Types of Algorithms

Usually depends on models, e.g. backrop for NN

Some generic schemes apply to many models:

- Generate and score
- Greedy search
- Genetic algorithms

Course Part II Summary

1. Planning: PDDL, POP, deterministic, observable
2. Uncertainty: Utilility and Probability
3. Graphical probability models (Bayes Nets)
    1. Semantics/distributions
    2. Inference in BNs

4. Decision-making U. uncertainty
    1. Preferences and utilities, MEU, VOI
    2. Sequential decision-making (MDP, a bit of POMDP)

5. Learning
    1. Decision trees
    2. Neural Networks
    3. Statistical + BNs
    4. Generalizations (not in needed material)